



NATIONAL CHEMISTRY CONTEST 2020

ANSWERS TO THE PROBLEMS

"It always seems impossible until it's done."

Nelson Mandela



Ministerio de Educación
República de Cuba



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Distribution of the problems by day and grade

	Day 1	Day 2
10mo	Problems 1, 2, 3 y 4	Problems 6 y 7
11no	Problems 1, 4 y 5	Problems 8 y 9
12mo	Problems 1, 4 y 5	Problems 10, 11 y 12

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Revision

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CONSTANTS, CONVERSIONS AND EQUATIONS

Avogadro's constant	$N_A = 6,0221 \cdot 10^{23} \text{ mol}^{-1}$	Ideal gas equation	$p \cdot V = n \cdot R \cdot T$
Gases contents	$R = 8,3145 \text{ J} \cdot \text{K}^{-1} \cdot \text{mol}^{-1}$	The third principle of thermodynamics	$Q = \Delta E + W$
Zeroth in the Celsius scale	273,15 K	Gibbs' energy	$\Delta G = \Delta H - T \cdot \Delta S$
Faraday's constant	$F = 96485 \text{ C} \cdot \text{mol}^{-1}$	Nernst's equation	$E = E^\circ - \frac{R \cdot T}{z \cdot F} \cdot \ln \frac{c_{\text{red}}}{c_{\text{ox}}}$
Ionic product of water at 25 °C	$K_W = 1,00 \cdot 10^{-14}$	Arrhenius' equations	$k = A \cdot \exp \left(-\frac{E_A}{R \cdot T} \right)$
Standard conditions	$p^\circ = 100 \text{ kPa}$ $c^\circ = 1 \text{ mol} \cdot \text{L}^{-1}$		$\Delta G^\circ = -R \cdot T \cdot \ln K^\circ = -z \cdot F \cdot \Delta E^\circ$
$1 \text{ J} = 1 \text{ C} \cdot 1 \text{ V} = 1 \text{ kPa} \cdot 1 \text{ L} = 1 \text{ W} \cdot 1 \text{ s}$		$1 \text{ C} = 1 \text{ A} \cdot 1 \text{ s}$	

PERIODIC TABLE OF THE ELEMENTS

PERIODIC TABLE OF THE ELEMENTS																		1
1 H 1.008	2											13	14	15	16	17	2 He 4.003	
3 Li 6.94	4 Be 9.01											5 B 10.81	6 C 12.01	7 N 14.01	8 O 16.00	9 F 19.00	10 Ne 20.18	
11 Na 22.99	12 Mg 24.31	3	4	5	6	7	8	9	10	11	12	13 Al 26.98	14 Si 28.09	15 P 30.97	16 S 32.06	17 Cl 35.45	18 Ar 39.95	
19 K 39.10	20 Ca 40.08	21 Sc 44.96	22 Ti 47.87	23 V 50.94	24 Cr 52.00	25 Mn 54.94	26 Fe 55.85	27 Co 58.93	28 Ni 58.69	29 Cu 63.55	30 Zn 65.38	31 Ga 69.72	32 Ge 72.63	33 As 74.92	34 Se 78.97	35 Br 79.90	36 Kr 83.80	
37 Rb 85.47	38 Sr 87.62	39 Y 88.91	40 Zr 91.22	41 Nb 92.91	42 Mo 95.95	43 Tc -	44 Ru 101.1	45 Rh 102.9	46 Pd 106.4	47 Ag 107.9	48 Cd 112.4	49 In 114.8	50 Sn 118.7	51 Sb 121.8	52 Te 127.6	53 I 126.9	54 Xe 131.3	
55 Cs 132.9	56 Ba 137.3	57-71	72 Hf 178.5	73 Ta 180.9	74 W 183.8	75 Re 186.2	76 Os 190.2	77 Ir 192.2	78 Pt 195.1	79 Au 197.0	80 Hg 200.6	81 Tl 204.4	82 Pb 207.2	83 Bi 209.0	84 Po -	85 At -	86 Rn -	
87 Fr -	88 Ra -	89-103	104 Rf -	105 Db -	106 Sg -	107 Bh -	108 Hs -	109 Mt -	110 Ds -	111 Rg -	112 Cn -	113 Nh -	114 Fl -	115 Mc -	116 Lv -	117 Ts -	118 Og -	

57 La 138.9	58 Ce 140.1	59 Pr 140.9	60 Nd 144.2	61 Pm -	62 Sm 150.4	63 Eu 152.0	64 Gd 157.3	65 Tb 158.9	66 Dy 162.5	67 Ho 164.9	68 Er 167.3	69 Tm 168.9	70 Yb 173.0	71 Lu 175.0
89 Ac -	90 Th 232.0	91 Pa 231.0	92 U 238.0	93 Np -	94 Pu -	95 Am -	96 Cm -	97 Bk -	98 Cf -	99 Es -	100 Fm -	101 Md -	102 No -	103 Lr -

Problem 1. Miscellanea (15 points)

1.1	...	Total	Total
1	idem	15	15
1	idem	15	15

Select the right answer in each case.

1.1. The air contains 21,0% in volume of O₂. The percent of methane (CH₄) that would be necessary in the atmosphere to consume all the O₂ through complete combustion is:

A) 21,0%	B) 10,5%	C) 42,0%	D) cannot be calculated
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1.2. Calcium salts provide the flame with color:

A) orange	B) magenta	C) brick red	D) purple
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1.3. The electronic structure of manganese is:

A) [Ar]3d ⁷	B) [Ar]3d ⁶ 4s ¹	C) [Ar]3d⁵4s²	D) [Ar]3d ⁴ 4s ² 4p ¹
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1.4. It has the highest melting point:

A) Na	B) Li	C) K	D) Cs
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1.5. Substances with the same chemical bond:

A) N₂, Na₂, H₂	B) Au, CsAu, NaK	C) O ₂ , SO ₃ , ZnO ₂	D) FeO, Al ₂ O ₃ , SiO ₂
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1.6. It has the lowest boiling point:

A) I ₂	B) Br ₂	C) Cl ₂	D) F₂
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1.7. They are chemical species with the same valence shell electron count:

A) N ₃ ⁻ , CON, COF	B) PH₃, H₂O, HF	C) SF ₄ , SOCl ₂ , IF ₃	D) CO ₂ , NO ₂ ⁻ , NO ₂ ⁺
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1.8. It has the highest secondary ionization energy:

A) Ca	B) He	C) O	D) Na
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1.9. It has a high ion polarization:

A) CsF	B) MgO	C) AuI₃	D) Na ₂ S
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1.10. It is the least water-soluble substance:

A) CH ₃ CH ₂ OH	B) Na	C) HCl	D) SiO₂
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1.11. The larger ion in the series P³⁻, P⁻, S²⁻, S⁻, Cl⁻ is:

A) S ⁻	B) P ⁻ ,	C) Cl ⁻	D) P³⁻
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1.12. It has a very small electron affinity:

A) Be	B) B	C) K	D) Ga
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1.13. It has the highest density:

A) Os	B) Cs	C) Hg	D) Pb
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1.14. It possesses the smallest electric conductivity:

A) Pt	B) Si	C) H₂O	D) Hg
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1.15. The oxidation number(s) of sulfur atoms in thiosulfate ion, S₂O₃²⁻, is(are):

A) 1 y 3+	B) 2+	C) 3+	D) 0 y 4+
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Problem 2. Chemical reactions (10 points)**[ONLY FOR 10th GRADE]**

2.1	...	Total	Total
1	idem	20	10
1	idem	20	10

Complete and balance the following chemical equations. Take into account the aggregation states. Indicate when no reaction happens.

- 2.1. $\text{FeCl}_3(\text{ac}) + 3\text{KOH}(\text{ac}) = \text{Fe}(\text{OH})_3(\text{s}) + 3\text{KCl}(\text{ac})$
- 2.2. $\text{CuSO}_4(\text{ac}) + 4\text{NaOH}(\text{ac, exc.}) = \text{Na}_2[\text{Cu}(\text{OH})_4](\text{ac}) + \text{Na}_2\text{SO}_4(\text{ac})$
- 2.3. $\text{NH}_4\text{Cl}(\text{ac}) + \text{NaOH}(\text{ac, conc.}) \triangleq \text{NH}_3(\text{g}) + \text{H}_2\text{O}(\text{l}) + \text{NaCl}(\text{ac})$
- 2.4. $\text{NH}_4\text{Al}(\text{SO}_4)_2(\text{ac}) + 2\text{Ba}(\text{OH})_2(\text{ac}) = \text{NH}_3(\text{ac}) + \text{Al}(\text{OH})_3(\text{s}) + 2\text{BaSO}_4(\text{ac}) + \text{H}_2\text{O}(\text{l})$
- 2.5. $\text{Fe}_2\text{O}_3(\text{s}) + 2\text{Al}(\text{s}) \triangleq \text{Al}_2\text{O}_3(\text{s}) + 2\text{Fe}(\text{s})$
- 2.6. $\text{CaC}_2(\text{s}) + 2\text{H}_2\text{O}(\text{l}) = \text{Ca}(\text{OH})_2(\text{ac}) + \text{C}_2\text{H}_2(\text{g})$
- 2.7. $\text{Al}(\text{s}) + 3\text{H}_2\text{O}(\text{l}) + \text{NaOH}(\text{ac}) \triangleq \text{Na}[\text{Al}(\text{OH})_4](\text{ac}) + 3/2\text{H}_2(\text{g})$
- 2.8. $\text{PCl}_3(\text{l}) + 3\text{H}_2\text{O}(\text{l}) = \text{H}_3\text{PO}_3(\text{ac}) + 3\text{HCl}(\text{ac})$
- 2.9. $\text{FeO}(\text{s}) + \text{H}_2(\text{g}) \triangleq \text{Fe}(\text{s}) + \text{H}_2\text{O}(\text{g})$
- 2.10. $\text{Mg}(\text{s}) + \text{H}_2\text{SO}_4(\text{ac}) = \text{MgSO}_4(\text{ac}) + \text{H}_2(\text{g})$
- 2.11. $3\text{H}_2\text{SO}_4(\text{ac}) + 2\text{Fe}(\text{OH})_3(\text{s}) = \text{Fe}_2(\text{SO}_4)_3(\text{ac}) + 6\text{H}_2\text{O}(\text{l})$
- 2.12. $\text{NaHSO}_4(\text{ac}) + \text{Na}_2\text{CO}_3(\text{ac}) = \text{Na}_2\text{SO}_4(\text{ac}) + \text{NaHCO}_3(\text{ac})$
- 2.13. $\text{C}(\text{s}) + \text{O}_2(\text{g, exc.}) = \text{CO}_2(\text{g})$
- 2.14. $\text{Ba}(\text{s}) + \text{O}_2(\text{s}) = \text{BaO}_2(\text{s})$
- 2.15. $\text{Mg}(\text{s}) + \text{N}_2(\text{g}) \triangleq \text{Mg}_2\text{N}_3(\text{s})$
- 2.16. $\text{CO}_2(\text{g}) + \text{Ca}(\text{OH})_2(\text{ac}) = \text{CaCO}_3(\text{s}) + \text{H}_2\text{O}(\text{l})$
- 2.17. $\text{Cu}(\text{s}) + 4\text{HNO}_3(\text{ac, conc.}) \triangleq \text{Cu}(\text{NO}_3)_2(\text{ac}) + 2\text{NO}_2(\text{g}) + 2\text{H}_2\text{O}(\text{l})$
- 2.18. $\text{Au}(\text{s}) + \text{HCl}(\text{ac}) \triangleq \text{no reaction}$
- 2.19. $2\text{O}_3(\text{g}) \triangleq 3\text{O}_2(\text{g})$
- 2.20. $\text{P}_4\text{O}_{10}(\text{s}) + 6\text{H}_2\text{O}(\text{l}) = 4\text{H}_3\text{PO}_4(\text{ac})$

exc.: excess reagent; conc.: highly concentrated

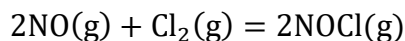
1 mark / complete equation (all correct products)

A penalty of -0,25 marks for each incorrect aggregation state. A penalty of -0,5 marks for each unbalanced equation. The penalty cannot exceed the value of the specific item.

Problem 3. Kinetics (10 points)**[ONLY FOR 10th GRADE]**

4.1	4.2	4.3	4.4	4.5	11.6	Total	Total
4	3	5	2	3	3	20	10
4	3	5	2	3	3	20	10

The reaction between nitric oxide and chlorine gives nitrosyl chloride as result. At 27 °C the following kinetic data was obtained.



Experiment	Initial concentration ($\text{mol}\cdot\text{L}^{-1}$)		Initial rate ($\text{mol}\cdot\text{L}^{-1}\cdot\text{s}^{-1}$)
	NO	Cl_2	
1	0,020	0,010	$8,27\cdot 10^{-5}$
2	0,020	0,020	$1,65\cdot 10^{-4}$
3	0,020	0,040	$3,31\cdot 10^{-4}$
4	0,040	0,020	$6,60\cdot 10^{-4}$
5	0,010	0,020	$4,10\cdot 10^{-5}$

3.1. Find the rate law for the equation.

$$v = k_r \cdot c^a(\text{NO}) \cdot c^b(\text{Cl}_2)$$

$$\frac{\text{Exp 1}}{\text{Exp 2}} = \frac{8,27 \cdot 10^{-5}}{1,65 \cdot 10^{-4}} = \frac{k_r \cdot 0,020^a \cdot 0,010^b}{k_r \cdot 0,020^a \cdot 0,020^b}$$

$$0,501 = 0,5^b \rightarrow b = 1$$

1 mark

$$\frac{\text{Exp 2}}{\text{Exp 4}} = \frac{1,65 \cdot 10^{-4}}{6,60 \cdot 10^{-4}} = \frac{k \cdot 0,020^a \cdot 0,020^b}{k \cdot 0,040^a \cdot 0,020^b}$$

$$0,25 = 0,5^a \rightarrow a = 2$$

1 mark

$$v = k_r \cdot c^2(\text{NO}) \cdot c(\text{Cl}_2)$$

2 marks

A correct non-numerical answer is also accepted.

4 marks

3.2. Calculate the average value of the rate constant.

$$k_r = \frac{v}{c^2(\text{NO}) \cdot c(\text{Cl}_2)}$$

$$\text{Exp 1} = 20,675 \quad \text{Exp 2} = 20,625 \quad \text{Exp 3} = 20,6875 \quad \text{Exp 4} = 20,625 \quad \text{Exp 2} = 20,5$$

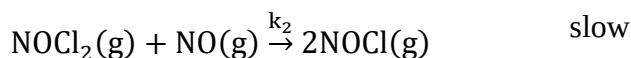
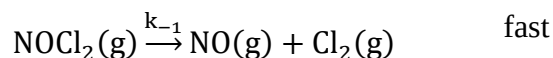
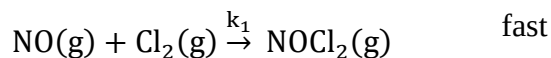
$$k_r = 20,6 \text{ L} \cdot \text{mol}^{-2} \cdot \text{s}^{-2}$$

0,5 marks c/u 0,5 marks / measure units -0,5 marks if less different than 3 significant figures

A correct non-averaged rate constant is accepted.

3 marks

It is possible that the reaction proceeds according to the mechanism



3.3. Use the steady-state approximation and find the rate law form the proposed mechanism.

The reaction rate depends mainly on the slow step

$$v = k_2 \cdot c(\text{NOCl}_2) \cdot c(\text{NO}) \quad 1 \text{ mark}$$

Assuming steady-state NOCl_2

$$\frac{dc(\text{NOCl}_2)}{dt} = 0 = k_1 \cdot c(\text{NO}) \cdot c(\text{Cl}_2) - k_{-1} \cdot c(\text{NOCl}_2) - k_2 \cdot c(\text{NOCl}_2) \cdot c(\text{NO}) \quad 2 \text{ marks}$$

$$\frac{k_1 \cdot c(\text{NO}) \cdot c(\text{Cl}_2)}{k_{-1} + k_2 \cdot c(\text{NO})} = c(\text{NOCl}_2) \quad 1 \text{ mark}$$

$$v = k_2 \cdot c(\text{NOCl}_2) \cdot c(\text{NO}) = \frac{k_2 \cdot k_1 \cdot c(\text{NO}) \cdot c(\text{Cl}_2) \cdot c(\text{NO})}{k_{-1} + k_2 \cdot c(\text{NO})}$$

$$v = \frac{k_2 \cdot k_1 \cdot c^2(\text{NO}) \cdot c(\text{Cl}_2)}{k_{-1} + k_2 \cdot c(\text{NO})} \quad 1 \text{ mark}$$

Other approaches conducting to the same result are also accepted.

5 marks

3.4. Under what conditions the previous expression (item 3.3) matches the experimental rate law (item 3.1)?

$$v = k_r \cdot c^2(\text{NO}) \cdot c(\text{Cl}_2) \text{ when } k_{-1} \gg k_2 \cdot c(\text{NO}).$$

This happens when the nitrogen monoxide concentration is low.

2 marks

3.5. It is known that the activation energy of the reaction is $22 \text{ kJ} \cdot \text{mol}^{-1}$. Calculate the rate constant when the temperature increases 10°C . If you cannot solve item 3.2 you may consider that $k_r = 50 \text{ u}$, where u symbolizes the measure units of the constant.

$$\ln\left(\frac{k'}{k}\right) = -\frac{E_A}{R} \left(\frac{1}{T_2} - \frac{1}{T_1}\right) \quad 1 \text{ mark}$$

$$\ln\left(\frac{k'}{20,6}\right) = -\frac{22000}{8,314}\left(\frac{1}{310} - \frac{1}{300}\right)$$

1 mark

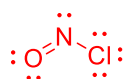
$$k' = 27,4 \text{ L} \cdot \text{mol}^{-2} \cdot \text{s}^{-2}$$

1 mark

When $k_r = 50 \text{ u}$ is used the result is $k_r' = 66 \text{ u}$.

3 marks

3.6. Represent the Lewis' formula of NOCl. Classify this molecule according to its polarity.



2 marks

polar molecule

1 mark

3 marks

Problem 4. Stoichiometric calculations¹ (15 points)

4.1	4.2	4.3	4.4	4.5	Total	Total
7	4	9	4	6	30	15
7	4	9	4	6	30	15

A certain metallic oxide contains 22,55% mass of oxygen. Another oxide of the same metal contains 50,48% mass of oxygen.

4.1. What is the relative atomic mass of the metal?

In M_2O_x is valid that

$$\frac{2}{x} = \frac{\omega_{\%}(M)}{A_r(M)} : \frac{\omega_{\%}(O)}{A_r(O)} = \frac{77,45}{A_r(M)} : \frac{22,55}{16,00} = \frac{54,95}{A_r(M)} \quad (1) \quad 2 \text{ marks}$$

In M_2O_y is valid that

$$\frac{2}{y} = \frac{\omega_{\%}(M)}{A_r(M)} : \frac{\omega_{\%}(O)}{A_r(O)} = \frac{49,52}{A_r(M)} : \frac{50,48}{16,00} = \frac{15,695}{A_r(M)} \quad (2) \quad 2 \text{ marks}$$

Dividing both expressions one finds

$$\frac{y}{x} = \frac{54,95}{15,695} = 3,5 = \frac{7}{2} \quad 1 \text{ mark}$$

Substituting $x = 2$ in expression (1)

$$A_r(M) = 54,95 \text{ g} \cdot \text{mol}^{-1} \rightarrow M = \text{Mn} \quad 2 \text{ marks}$$

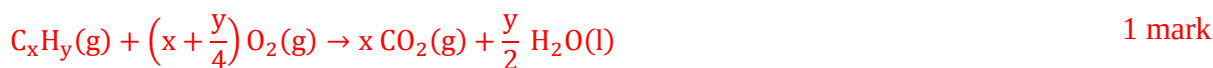
Other correct ways of solving are accepted, for example, trial and error. 2 marks

4.2. Find the empirical formula of the oxides.

MnO 2 marks Mn_2O_7 2 marks 4 marks

10 mL of an unknown gaseous hydrocarbon were mixed with 70 mL oxygen and allowed to react by an electric spark. When the reaction finished and all water vapor formed was condensed, the final volume of the gases was 65 mL. When the gaseous mixture reacted with a solution of potassium hydroxide, the volume decreased until 45 mL.

4.3. Find the formula of the unknown hydrocarbon if all the gas volumes were measure at TPN.



$$n(O_2) \text{ that reacted} = n_i(O_2) - n_f(O_2) = \frac{70}{22,4} - \frac{45}{22,4} = \frac{25}{22,4} \text{ mol} \quad 1 \text{ mark}$$

¹ International Chemistry Olympiad 1969, problem 3. International Chemistry Olympiad 1970, problems 4 and 6.

$$n(\text{CO}_2) = v_f(\text{total}) - n_f(\text{O}_2) = \frac{65}{22,4} - \frac{45}{22,4} = \frac{20}{22,4} \text{ mol} = n(\text{O}_2) \text{ que forma CO}_2 \quad 1 \text{ mark}$$

$$\text{Then, } n(\text{O}_2) \text{ that forms water} = \frac{25}{22,4} - \frac{20}{22,4} = \frac{5}{22,4} \text{ mol} \quad 1 \text{ mark}$$

$$\text{Therefore, } n(\text{H}_2\text{O}) = \frac{10}{22,4} \text{ mol} \quad 1 \text{ mark}$$

$$x = \frac{n(\text{CO}_2)}{n(\text{C}_x\text{H}_y)} = \frac{20}{10} = 2 \quad 1 \text{ mark}$$

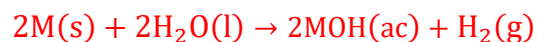
$$\frac{y}{2} = \frac{n(\text{H}_2\text{O})}{n(\text{C}_x\text{H}_y)} = \frac{10}{10} = 1 \rightarrow y = 2 \quad 1 \text{ mark}$$

$$\text{C}_x\text{H}_y = \text{C}_2\text{H}_2 \quad 2 \text{ marks}$$

Alternative solutions are accepted, for example, using directly the volume values 9 marks instead of the amount of substance.

An alloy is formed by rubidium mixed with other alkaline metals. A sample of 4,60 g of the alloy was reacted with water, releasing 2,241 L of hydrogen gas, with volume measured at normal temperature and pressure (TPN).

4.4. Which is the other metal composing the alloy?



$$n(\text{H}_2) = \frac{V(\text{O}_2)}{V_m} = \frac{2,241}{22,4} = 0,100 \text{ mol} \quad 1 \text{ mark}$$

$$n(\text{M}) = 2 \cdot n(\text{H}_2) = 2 \cdot 0,100 = 0,200 \text{ mol} \quad 1 \text{ mark}$$

$$M(\text{alloy}) = \frac{m(\text{alloy})}{n(\text{alloy})} = \frac{4,60}{0,200} = 23 \text{ g} \cdot \text{mol}^{-1} \quad 1 \text{ mark}$$

The only possibility is that the other metal is lithium. 1 mark 4 marks

4.5. What is the composition of the alloy in mass percent?

$$M(\text{Rb}) + m(\text{Li}) = 4,60$$

$$n(\text{Rb}) \cdot A_r(\text{Rb}) + n(\text{Li}) \cdot A_r(\text{Li}) = 4,60$$

$$85,5 \cdot n(\text{Rb}) + 7 \cdot n(\text{Li}) = 4,60 \quad \left. \begin{array}{l} \\ \\ \end{array} \right\} \quad 1 \text{ mark}$$

$$n(\text{Rb}) + n(\text{Li}) = 0,2 \quad \left. \begin{array}{l} \\ \\ \end{array} \right\} \quad 1 \text{ mark}$$

Solving the equation system, one gets

$$n(\text{Rb}) = 0,0408 \text{ mol} \rightarrow m(\text{Rb}) = 3,488 \text{ g} \quad 1 \text{ mark}$$

$$n(\text{Li}) = 0,1592 \text{ mol} \rightarrow m(\text{Li}) = 1,114 \text{ g} \quad 1 \text{ mark}$$

$$\omega(\text{Rb}) = \frac{m(\text{Rb})}{m(\text{aleación})} \cdot 100 = \frac{3,488}{4,60} \cdot 100 \approx 76\% \quad 1 \text{ mark}$$

$$\omega(\text{Li}) = \frac{m(\text{Li})}{m(\text{aleación})} \cdot 100 = \frac{1,114}{4,60} \cdot 100 \approx 24\% \quad 1 \text{ mark}$$

Other equivalent equation systems are accepted. 6 marks

$$V_{\text{m}}(\text{TPN}) = 22,4 \text{ L} \cdot \text{mol}^{-1}$$

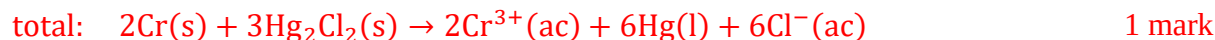
$$K = \exp\left(\frac{z \cdot F \cdot \Delta E^\circ}{R \cdot T}\right) = \exp\left(\frac{6 \cdot 96485 \cdot 0,74}{8,314 \cdot 298}\right) = 1,1 \cdot 10^{75}$$

2 marks

5 marks

A galvanic cell was built aiming to study the dependency of the electrode potential of the redox couple $\text{Cr}^{3+}(\text{ac})/\text{Cr}(\text{s})$ with the pH value. A calomel electrode saturated with potassium chloride (ECS, $\text{Hg}_2\text{Cl}_2(\text{s})/\text{Hg}(\text{l})$) was used as reference electrode. The working electrode was prepared with 50,0 mL of a 0,0100 M CrCl_3 aqueous solution and a chromium wire as electronic conductor. A solution saturated in KNO_3 functioned as salt bridge. Next, slow addition of small volumes of 0,0500 M NaOH solution were added until a total of 40,0 mL was added to the original solution. Finally, a plot of galvanic cell voltage as a function of the volume of added sodium hydroxide was build.

5.6. Write the half equations of cathode and anode and the total cell equation.



Balancing the anodic reaction in basic media is accepted. 3 marks

5.7. Mention three characteristics that the electrolyte in the salt bridge must meet.

The electrolyte in the salt bridge must be highly soluble and chemically inert. The molar conductivity of both ions must be high and similar. 2 marks

5.8. Estimate the initial emf of the cell. Neglect the hydrolysis of the Cr^{3+} ion.



$E(\text{Cr}^{3+}/\text{Cr}) = E^\circ - \frac{0,0592}{3} \log\left(\frac{1}{[\text{Cr}^{3+}]}\right)$ 1 mark

$E(\text{Cr}^{3+}/\text{Cr}) = -0,74 - \frac{0,0592}{3} \log\left(\frac{1}{0,01}\right) = -0,78 \text{ V}$ 2 marks

$\Delta E^\circ = E(\text{Hg}_2\text{Cl}_2/\text{Hg}) - E(\text{Cr}^{3+}/\text{Cr}(\text{s})) = 0,271 - (-0,78) = 1,05 \text{ V}$ 1 mark 5 marks

5.9. Estimate the final emf of the cell. Neglect the possible formation of complex ions



$n(\text{OH}^-) = 40 \cdot 0,05 = 2,00 \text{ mmol}$ 1 mark

$n(\text{Cr}^{3+}) = 50 \cdot 0,01 = 0,500 \text{ mmol}$ 1 mark

$c(\text{OH}^-) = \frac{2 - 3 \cdot 0,5}{40 + 50} = 5,56 \cdot 10^{-3} \text{ mol} \cdot \text{L}^{-1}$ 1 mark



$$E(\text{Cr(OH)}_3/\text{Cr}) = E^\circ - \frac{0,0592}{3} \log([\text{OH}^-]^3) \quad 1 \text{ mark}$$

$$E(\text{Cr(OH)}_3/\text{Cr}) = -1,33 - 0,0592 \cdot \log(5,56 \cdot 10^{-3}) = -1,20 \text{ V} \quad 1 \text{ mark}$$

$$\Delta E^\circ = E(\text{Hg}_2\text{Cl}_2/\text{Hg}) - E(\text{Cr(OH)}_3/\text{Cr}) = 0,271 - (-1,20) = 1,47 \text{ V} \quad 1 \text{ mark} \quad \underline{7 \text{ marks}}$$

Chromium deposition is usually performed by electrolysis of a solution of chromic acid. The anode is constituted by an alloy not reacting under the conditions employed, while the metallic object that should be coated works as cathode. An electrolytic cell was filled with 100 L of an aqueous solution containing 0,250 kg/L of CrO_3 . During the electrolysis, a current of 1500 A was applied for 10,0 h. The efficiency of the process was 15%.

5.10. Calculate the mass of chromium deposited in the cathode.



$$m(\text{Cr})_{\text{teo}} = \frac{i \cdot t \cdot A_r(\text{Cr})}{z \cdot F} = \frac{1500 \cdot 10 \cdot 3600 \cdot 52,00}{6 \cdot 96485} = 4850 \text{ g} \quad 2 \text{ marks}$$

$$m(\text{Cr}) = 0,15 \cdot 4850 = 728 \text{ g} \quad 1 \text{ mark}$$

Other half equations for the reduction to metallic chromium are acceptable. 4 marks

5.11. Write the half equations corresponding to two collateral redox processes that might happen in the previous electrolytic cell and prove that the efficiency is lower than 100%.



Par redox	$\text{Hg}_2\text{Cl}_2(\text{s})/\text{Hg}(\text{l}), \text{KCl}(\text{sat})$	$\text{O}_2(\text{g})/\text{H}_2\text{O}(\text{l})$	$\text{H}^+(\text{ac})/\text{H}_2(\text{g})$
E°/V	0,271 V	1,23	0,000

Problem 6. Hydrogen compounds (30 points)

6.1	6.2	6.3	6.4	6.5	6.6	6.7	6.8	6.9	6.10
2	6	4	3	2	4	4	2	5	2
2	6	4	3	2	4	4	2	5	2
6.11	6.12	6.13	6.14	6.15	6.16	6.17		Total	Total
4	2	2	7	3	4	4		60	30
4	2	2	7	3	4	4		60	30

Hydrogen is the lightest and most abundant element in the universe. At standard temperature and pressure, it exists in the molecular form H_2 . This gas is colorless, odorless, insoluble in water and flammable. In the laboratory, hydrogen can be obtained by the reaction between zinc and hydrochloric acid at room temperature, or by heating aluminum or silicon with sodium hydroxide.

6.1. Hydrogen element exists in the form of proton 1H , deuterium 2H and tritium 3H , though other isotopes have been artificially produced. What is the difference between these atoms?

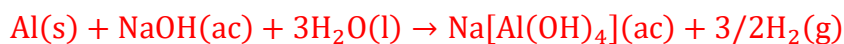
They have different number of electrons.

3 marks

6.2. Formulate the chemical reactions for the generation of hydrogen in the laboratory.



2 marks



2 marks



2 marks

–0,5 marks for an incorrect aggregation state. –1 mark if the equation is not balanced. The penalty cannot exceed the value of each equation.

6 marks

6.3. Compare the distance and bond energy in the hydrogen species H_2 , H_2^+ y H_2^- .

Bond distance: $H_2 < H_2^+ \approx H_2^-$

2 marks

Bond energy: $H_2 > H_2^+ \approx H_2^-$

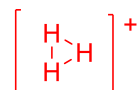
2 marks

It is accepted to say that H_2^- has the largest bond distance and the smallest bond energy of all.

4 marks

Ion H_3^+ has been detected. It is a highly unstable species, diamagnetic, having a very symmetric structure, in which three atoms are found the closest possible.

6.4. Propose a structure for H_3^+ , showing the special distribution of all atoms; it is not necessary to represent the bonds. How many occupied molecular orbitals present this ion?



2 marks

One MO is occupied.

1 mark

3 marks

Hydrogen forms binary compounds with many elements of the periodic table. These hydrides can be classified according to the chemical bond and the structure that they have in saline (group 1 and 2),

metallic (some transition and internal transition elements), intermediate (Cu and Zn) and molecular (Be and groups 13-17).

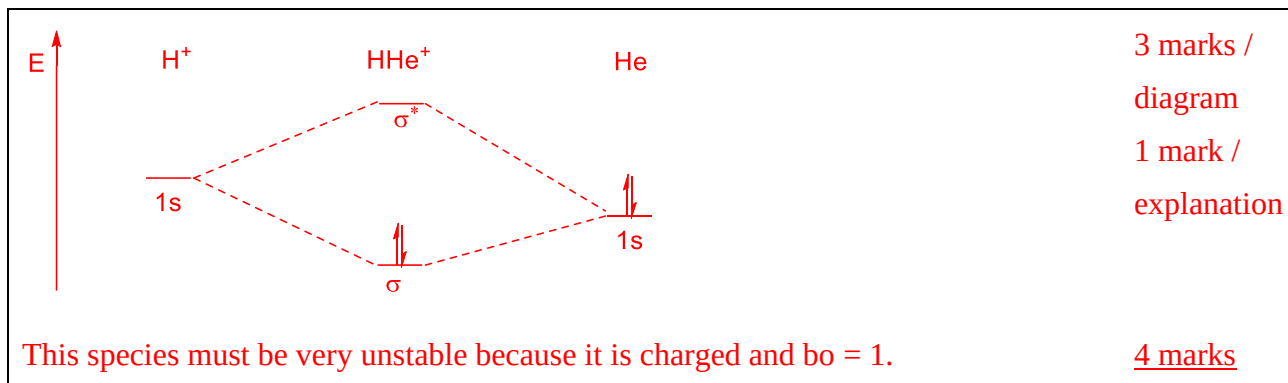
6.5. Which is the chemical bond predominant in saline hydrides and molecular hydrides?

saline hydride: ionic bond

Molecular hydride: covalent bond

2 marks

6.6. Build an approximate energetic diagram of molecular orbital for the ion-molecule of HHe^+ . Why can we predict that this molecule is unstable?

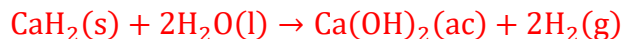


Calcium hydride is used as a portable hydrogen generator in remote places, for example, for filling meteorological balloons. It is convenient because it is cheap and reacts with water at room temperature.

6.7. Write the chemical equation for the obtention of calcium hydride and its reaction with water.



2 marks



2 marks

–0,5 marks for an incorrect aggregation state. –1 mark if the equation is non-balanced. The penalty cannot exceed the value for the equation.

4 marks

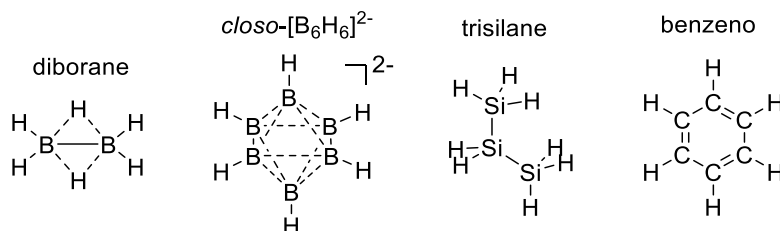
Numerous molecular hydrides are presented in polymeric form. The best examples of this property are the millions of organic molecules that exist.

6.8. How do you explain that carbon forms four chemical bonds in the organic molecules, if the isolated atoms only possess two unpaired electrons?

All valence orbitals of carbon participate in the bond formation.

2 marks

6.9. Draw the electronic structure of ethyne according to the Valence Bond Theory. Use the hybridization model for the carbon atom.



Boron combines with hydrogen to form boranes, among which the simplest compounds is B₂H₆. In borane molecules the electrons of some hydrogen or boron atoms are shared between three or more atoms. The “unusual” covalent chemical bond is because their molecules are electron deficient, and therefore, cannot complete an octet around boron atoms.

6.10. Why are boranes electron deficient molecules?

Boron atom has three electrons in their last energy level.

2 marks

6.11. Formulate the chemical equations for the obtention of mixed hydrides LiAlH₄ and NaBH₄ for the corresponding binary hydrides.



2 marks



2 marks



–0,5 marks for an incorrect aggregation state. –1 mark if the equation is not balance. 4 marks
The penalty cannot exceed the value for each equation.

The tendency to form polymeric molecules diminishes when going down a group in the periodic table. Carbon is able to virtually form infinite chains in its hydrogen compounds, while the largest analogous silane only contains seven silicon atoms (Si₇H₁₆). On one hand hydrocarbons are stable at room temperature and a heat source is needed to start combustion; on the other hand, SiH₄ spontaneously ignites in the air.

6.12. Mention one factor causing that the silanes are least stable and more reactive than the hydrocarbons.

Because the silicon atom has a larger size the chemical bonds Si-Si and Si-H are weaker. / Silicon possesses empty d orbitals which are able to accept electrons. / Silicon can form strong bonds with very electronegative atoms such as F, O and N. 2 marks

The boiling point of molecular hydrides for groups 14-17 generally increase while going down the group. The exception to this rule is hydrides H_2O , NH_3 and HF , which possess the highest boiling point of the group they belong to.

6.13. Why H_2O , NH_3 and HF molecules have high boiling points?

H_2O , NH_3 and HF molecules have hydrogen bonds. Therefore, for boiling to occur it is necessary an additional energy to break these bonds. 2 marks

6.14. Draw the Lewis' structures for CH_4 , NH_3 , H_2O and HF . Predict the molecular geometry and the bond angle for the first three of them.

$\begin{array}{c} \text{H} \\ | \\ \text{H}-\text{C}-\text{H} \\ | \\ \text{H} \end{array}$
 tetrahedral 109°
 $\begin{array}{c} \text{H} \\ | \\ \text{:N}-\text{H} \\ | \\ \text{H} \end{array}$
 trigonal 120°
 $\begin{array}{c} \text{H} \\ | \\ \text{:O}-\text{H} \\ | \\ \text{:} \end{array}$
 angular 109°
 $\text{H}-\ddot{\text{F}}\text{:}$

1 mark / structure 0,5 marks / geometry or angle 7 marks

Besides ammonia NH_3 , there are numerous binary compounds formed by nitrogen and hydrogen, such as hydrazoic acid HN_3 , hydrazine N_2H_4 and diimide or diazene N_2H_2 .

6.15. Which is the average oxidation state for nitrogen in the previous molecules?

HN_3 $-1/3$ N_2H_4 -2 N_2H_2 -1 3 marks

6.16. Draw resonant structures for azide ion N_3^- which is formed when dissolving hydrazoic acid in water. Which is the most contributing resonance structure?

$\left[\begin{array}{c} \ddot{\text{N}}=\ddot{\text{N}}=\ddot{\text{N}} \\ \text{most contributing} \end{array} \longleftrightarrow \begin{array}{c} \text{:N}\equiv\text{N}-\ddot{\text{N}}\text{:} \\ \text{:} \end{array} \longleftrightarrow \begin{array}{c} \ddot{\text{N}}-\text{N}\equiv\text{N}\text{:} \\ \text{:} \end{array} \right]^-$

1 mark / structure 1 mark / most contributing 4 marks

Group 17 hydrides are formed by direct reaction between hydrogen and halogen at high temperatures. There are indirect obtention methods; for example, hydrogen chloride can be obtained by the reaction between solid sodium chloride and concentrated sulfuric acid.

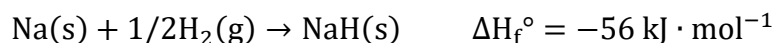
6.17. Formulate the chemical equations corresponding to the direct and indirect obtention methods for hydrogen chloride.

$\text{H}_2(\text{g}) + \text{H}_2(\text{g}) \rightarrow 2\text{HCl}(\text{g})$ 2 marks
 $\text{NaCl}(\text{s}) + \text{H}_2\text{SO}_4(\text{ac}) \rightarrow \text{HCl}(\text{g}) + \text{NaHSO}_4(\text{ac})$ 2 marks
 $-0,5$ marks for an incorrect aggregation state. -1 mark if the equation is unbalanced. 4 marks
 The penalty cannot exceed the value of the equation.

Problem 7. Hydrides in Thermochemistry (20 points)

7.1	7.2	7.3	7.4	7.5	Total	Total
7	8	8	8	9	40	20
7	8	8	8	9	40	20

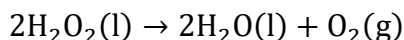
Sodium hydride is formed by the reaction between hydrogen and sodium.



7.1. Calculate the lattice enthalpy of this substance through the thermodynamic Born-Haber cycle.

$\Delta H_f^\circ(\text{NaH}) = \Delta H_{\text{sub}}^\circ(\text{Na}) + 0,5 \cdot \Delta H_{\text{dis}}^\circ(\text{H}_2) + I(\text{Na}) + A(\text{H}) + U(\text{NaH})$	3 marks
$-56 = 107 + 0,5 \cdot 436 + 496 - 73 + U(\text{NaH})$	2 marks
$U(\text{NaH}) = -804 \text{ kJ} \cdot \text{mol}^{-1}$	2 marks <u>7 marks</u>

Hydrogen peroxide is thermodynamically unstable and slowly decomposes forming water and oxygen. This process is accelerated in the presence of bases.



7.2. Given the values of mean bond dissociation enthalpies, calculate the approximate variation enthalpy for the previous reaction.

$\Delta H_r^\circ = \sum \Delta H^\circ(\text{broken bonds}) - \sum \Delta H^\circ(\text{bonds formed})$	2 marks
$\Delta H_r^\circ = 4 \cdot \Delta H^\circ(\text{O} - \text{H}) + 2 \cdot \Delta H^\circ(\text{O} - \text{O}) - 4 \cdot \Delta H^\circ(\text{O} - \text{H}) - \Delta H^\circ(\text{O} = \text{O})$	2 marks
$\Delta H_r^\circ = 2 \cdot 146 - 497$	2 marks
$\Delta H_r^\circ = -206 \text{ kJ} \cdot \text{mol}^{-1}$	2 marks <u>8 marks</u>

Methane is the majority component of natural gas. This hydrocarbon is produced by the putrefaction of organic matter, naturally or because of human action. Its principal application is as fuel.

7.3. Calculate the combustion enthalpy of methane, that is, the enthalpy variation corresponding to the reaction of methane with oxygen.

$\text{CH}_4(\text{g}) + 2\text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g}) + 2\text{H}_2\text{O}(\text{l})$	2 marks
$\Delta H_r^\circ = \sum v \cdot \Delta H_f^\circ$	1 mark
$\Delta H_c^\circ(\text{CH}_4) = \Delta H_f^\circ(\text{CO}_2) + 2 \cdot \Delta H_f^\circ(\text{H}_2\text{O}) - \Delta H_f^\circ(\text{CH}_4) - 2 \cdot \Delta H_f^\circ(\text{O}_2)$	2 marks
$\Delta H_c^\circ(\text{CH}_4) = -394 + 2 \cdot (-286) - (-75)$	1 mark
$\Delta H_c^\circ(\text{CH}_4) = -891 \text{ kJ} \cdot \text{mol}^{-1}$	2 marks <u>8 marks</u>

Part of the methane is found beneath the ocean in the form of combustible ice, which is formed by a clathrate of ideal composition $(\text{CH}_4)_8(\text{H}_2\text{O})_{46}$. These metal hydrides are formed at high pressure and low temperatures.

7.4. Which amount of energy can be obtained for the combustion of 1,00 g of methane hydrate? If you cannot solve item 7.3 consider that $\Delta H_{\text{comb}}^\circ(\text{CH}_4) = -900 \text{ kJ}\cdot\text{mol}^{-1}$.

$$M(\text{hydrate}) = 8 \cdot M(\text{CH}_4) + 46 \cdot M(\text{H}_2\text{O}) \quad 1 \text{ mark}$$

$$M(\text{hydrate}) = 8 \cdot 16,0 + 46 \cdot 18,0 = 956 \text{ g}\cdot\text{mol}^{-1} \quad 1 \text{ mark}$$

$$n(\text{hydrate}) = \frac{m(\text{hydrate})}{M(\text{hydrate})} = \frac{1,00}{956} = 0,00105 \text{ mol} \quad 2 \text{ marks}$$

$$n(\text{CH}_4) = 8 \cdot n(\text{hydrate}) = 8 \cdot 0,00105 = 0,00837 \text{ mol} \quad 2 \text{ marks}$$

$$\Delta H_r^\circ = n(\text{CH}_4) \cdot \Delta H_c^\circ(\text{CH}_4) = 0,00837 \cdot (-891) = -7,46 \text{ kJ} \quad 2 \text{ marks}$$

$$\text{Using the value of } -900 \text{ kJ/mol for } \Delta H_c^\circ(\text{CH}_4) \text{ the result is } -7,53 \text{ kJ.} \quad 8 \text{ marks}$$

One of the greater dangers because of global warming is that it may cause the release of methane retained in the bottom of the ocean and the permafrost. Methane is a greenhouse gas known with a more potent effect than CO_2 , in consequence, its release in large amounts at the atmosphere would further accelerate the heating effect. It is estimated that the total amount of methane in the earth is larger than $5 \cdot 10^{11}$ tons.

7.5. In how many degrees the earth atmosphere would be heated if that amount of methane is burnt with the atmospheric oxygen? The total heat capacity of the earth atmosphere has an approximate value of $4 \cdot 10^{21} \text{ J}\cdot\text{K}^{-1}$.

$$n(\text{CH}_4) = \frac{m(\text{CH}_4)}{M(\text{CH}_4)} = \frac{5 \cdot 10^{11} \cdot 10^6}{16,0} = 3,13 \cdot 10^{16} \text{ mol} \quad 2 \text{ marks}$$

$$\Delta H_r^\circ = n(\text{CH}_4) \cdot \Delta H_c^\circ(\text{CH}_4) \quad 1 \text{ mark}$$

$$\Delta H_r^\circ = 3,13 \cdot 10^{16} \cdot (-891) = -2,78 \cdot 10^{19} \text{ kJ} = -2,78 \cdot 10^{22} \text{ J} \quad 2 \text{ marks}$$

$$Q_{\text{atm}} = -\Delta H_r^\circ = 2,78 \cdot 10^{22} \text{ J} \quad 2 \text{ marks}$$

$$\Delta T = \frac{Q_{\text{atm}}}{C} = \frac{2,78 \cdot 10^{22}}{4,21 \cdot 10^{21}} = 6,96 \text{ K} \approx 7 \text{ K} \quad 2 \text{ marks}$$

$$\text{Using the value of } -900 \text{ kJ/mol for } \Delta H_c^\circ(\text{CH}_4) \text{ the same result is obtained.} \quad 9 \text{ marks}$$

$I(\text{H}) = 1312 \text{ kJ/mol}$, $I(\text{Na}) = 496 \text{ kJ/mol}$, $A(\text{H}) = -73 \text{ kJ/mol}$, $A(\text{Na}) = -53 \text{ kJ/mol}$, $\Delta H_{\text{dis}}^\circ(\text{H}_2) = 436 \text{ kJ/mol}$, $\Delta H_{\text{dis}}^\circ(\text{Na}_2) = 77 \text{ kJ/mol}$, $\Delta H_{\text{sub}}^\circ(\text{Na}) = 107 \text{ kJ/mol}$, $\Delta H_f^\circ(\text{NaH}) = -56 \text{ kJ/mol}$

Bond	H-O	O-O	O=O
$E_{\text{dis}}/\text{kJ}\cdot\text{mol}^{-1}$	463	146	497

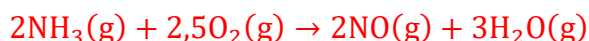
Substance	CH_4	H_2O	CO_2
$\Delta H_f^\circ/\text{kJ}\cdot\text{mol}^{-1}$	-75	-286	-394

Problem 8. Nitric acid and nitrogen oxides (25 points)

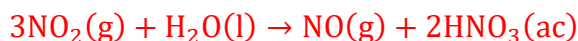
8.1	8.2	8.3	8.4	8.5	8.6	8.7	8.8	8.9	8.10	Total	Total
8	9	2	4	3	3	2	4	6	9	50	25
8	9	2	4	3	3	2	4	6	9	50	25

Nitric acid is obtained in a large scale by the Ostwald method. In this process, ammonia is first reacted with oxygen at high temperatures and in the presence of a Pt catalyst (with small amounts of Pd or Rh) forming nitric oxide (NO). Then, nitric oxide is oxidized to nitrogen dioxide. This oxide reacts with water forming nitric acid and nitric oxide. Regenerated NO is recovered and reincorporated in the productive cycle. The process becomes more expensive because a small amount of the platinum used is transformed in PtO_2 .

8.1. Write the equations corresponding to all the chemical reactions that happen in the Ostwald process.



2 marks c/u



A penalty of -1 mark for two incorrect aggregation states. A penalty of -1 mark if 8 marks the equation is non-balanced.

8.2. A commercial solution of nitric acid 65% mass has a density of $1,39 \text{ g} \cdot \text{mL}^{-1}$. Calculate i) the freezing point, ii) the boiling point and iii) the vapor pressure of the solution.

$$M(\text{HNO}_3) = 1,0 + 14,0 + 3 \cdot 16,0 = 63,0 \text{ g} \cdot \text{mol}^{-1}$$

$$M(\text{H}_2\text{O}) = 2 \cdot 1,0 + 16,0 + 16,0 = 18,0 \text{ g} \cdot \text{mol}^{-1}$$

$$b(\text{HNO}_3) = \frac{n(\text{HNO}_3)}{m(\text{H}_2\text{O})} = \frac{\omega_{\%}(\text{HNO}_3) \cdot 1000}{M(\text{HNO}_3)(100 - \omega_{\%})} \quad 1 \text{ mark}$$

$$b(\text{HNO}_3) = \frac{65 \cdot 1000}{63,0 \cdot (100 - 65)} = 29,5 \text{ mol} \cdot \text{kg}^{-1} \quad 1 \text{ mark}$$

$$\Delta T_c = i \cdot b(\text{HNO}_3) \cdot K_{cr} = 2 \cdot 29,5 \cdot 1,86 \approx 110 \text{ }^{\circ}\text{C} \quad 1 \text{ marks}$$

$$T_c = T_f - \Delta T_c = 0 - 110 = -110 \text{ }^{\circ}\text{C} \quad 1 \text{ marks}$$

$$\Delta T_{eb} = i \cdot b(\text{HNO}_3) \cdot K_{eb} = 2 \cdot 29,5 \cdot 0,52 \approx 31 \text{ }^{\circ}\text{C} \quad 1 \text{ marks}$$

$$T_{eb} = T_{eb} - \Delta T_{eb} = 100 + 31 = 131 \text{ }^{\circ}\text{C} \quad 1 \text{ marks}$$

$$p_D \approx X_d' \cdot p_d^{\circ} = \frac{n(\text{H}_2\text{O})}{n(\text{H}_2\text{O}) + i \cdot n(\text{HNO}_3)} \cdot p_d^{\circ} \quad 1 \text{ mark}$$

$$p_D = \frac{\frac{100 - \omega_{\%}(\text{HNO}_3)}{M(\text{H}_2\text{O})}}{\frac{100 - \omega_{\%}(\text{HNO}_3)}{M(\text{H}_2\text{O})} + \frac{2 \cdot \omega_{\%}(\text{HNO}_3)}{M(\text{HNO}_3)}} \cdot p_d^{\circ} = \frac{\frac{100 - 65}{18} \cdot 31,7}{\frac{100 - 65}{18} + \frac{2 \cdot 65}{63}} = 15,4 \text{ hPa}$$

2 marks

Other solutions are accepted, for example, assuming 100 g of solution. A penalty of 1 mark for no taking into account Van't Hoff's i factor. 9 marks

8.3. The previous reaction freezes at -32°C , boils at 121°C and has a vapor pressure of 9,4 hPa. Why are there differences between the experimental values and the values calculated by you?

The concentration is high, hence, there are strong inter ionic interactions and the solution does not behave ideally. 2 marks

8.4. What volume of commercial solution of nitric acid is needed to prepare 1,00 L of solution with concentration $0,50 \text{ mol}\cdot\text{L}^{-1}$?

$$c(\text{HNO}_3) = \frac{\omega_{\%}(\text{HNO}_3) \cdot 10 \cdot \rho_D}{M(\text{HNO}_3)} = \frac{65 \cdot 10 \cdot 1,39}{63} = 14,3 \text{ mol}\cdot\text{L}^{-1}$$

$$V(\text{conc}) = \frac{V(\text{dil}) \cdot c(\text{dil})}{c(\text{conc})} = \frac{1,00 \cdot 0,50}{14,3} = 0,035 \text{ L} = 35 \text{ mL}$$

2 marks

1 mark

3 marks

Nitric oxides are generated in small amounts as byproducts of the internal combustion engines and in factories at high room temperatures. They are powerful atmospheric pollutants due to their oxidizing power and because they generate acid rain.

8.5. Calculate the pH of acid rain that contains $3,2 \cdot 10^{-5} \text{ mol}\cdot\text{L}^{-1}$ of nitric acid.

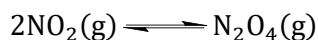
$$[\text{H}^+] = c(\text{HNO}_3) = 3,2 \cdot 10^{-5} \text{ mol}\cdot\text{L}^{-1}$$

$$\text{pH} = -\log(3,2 \cdot 10^{-5}) \approx 4,50$$

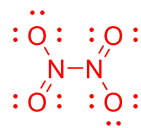
2 marks

2 marks 4 marks

Nitrogen dioxide exists in equilibrium with the N_2O_4 dimer, dinitrogen tetroxide. This oxide is used as rocket fuel.



8.6. Draw the Lewis structure of N_2O_4 .

3 marks

8.7. The conversion of NO_2 to N_2O_4 is favored at low temperatures. Why?

The formation of the N-N bond implies that ΔH_f° is negative for the formation of the dimer and, therefore, the equilibrium constant increases with decreasing temperature. 2 marks

To study the balance between these oxides, 2.00 mol of NO_2 were introduced into a 5.00 L rigid container, previously evacuated. The container was cooled with ice to 0°C .

8.8. Estimate the approximate total pressure (in kPa) inside the container at 0°C . Assume that only N_2O_4 dimer is present under these conditions.

Since the equilibrium constant is very large, all the NO_2 is transformed into N_2O_4 . 1 mark

$$n(\text{N}_2\text{O}_4) = 0,5 \cdot n(\text{NO}_2) = 0,5 \cdot 2,00 = 1,00 \text{ mol} \quad 1 \text{ mark}$$

$$p(\text{total}) = p(\text{N}_2\text{O}_4) = \frac{n(\text{N}_2\text{O}_4) \cdot R \cdot T}{V(\text{total})} = \frac{1,00 \cdot 8,314 \cdot 273,15}{5,00} = 454 \text{ kPa} \quad \begin{array}{l} 2 \text{ marks} \\ 4 \text{ marks} \end{array}$$

8.9. At what temperature is it true that $K_p^\circ = 1.00$ for the chemical equilibrium analyzed?



$$\Delta H_r^\circ = \Delta H_f^\circ(\text{N}_2\text{O}_4) - 2 \cdot \Delta H_f^\circ(\text{NO}_2) = 9,16 - 2 \cdot 33,18 = -57,2 \text{ kJ} \cdot \text{mol}^{-1} \quad 1 \text{ mark}$$

$$\Delta S_r^\circ = S^\circ(\text{N}_2\text{O}_4) - 2 \cdot S^\circ(\text{NO}_2) = 304,29 - 2 \cdot 240,06 = -175,83 \text{ J} \cdot \text{K}^{-1} \cdot \text{mol}^{-1} \quad 1 \text{ mark}$$

$$\Delta G_r^\circ = -R \cdot T \cdot \ln K_p^\circ = -8,314 \cdot T \cdot \ln 1,00 = 0 \quad 1 \text{ mark}$$

$$\Delta G_r^\circ = \Delta H_r^\circ - T \cdot \Delta S_r^\circ = 0 \quad 1 \text{ mark}$$

$$T = \frac{\Delta H_r^\circ}{\Delta S_r^\circ} = \frac{-57200}{-175,83} = 325 \text{ K} = 52^\circ\text{C} \quad \begin{array}{l} 2 \text{ marks} \\ 6 \text{ marks} \end{array}$$

8.10. Calculate the total pressure (in kPa) inside the container at the above temperature. If you could not calculate the value of the temperature in 8.8, use the value $T = 50^\circ\text{C}$.

$$p_i(\text{NO}_2) = \frac{n(\text{NO}_2) \cdot R \cdot T}{V} = \frac{2 \cdot 8,314 \cdot 325}{5,0} = 1,081 \cdot 10^3 \text{ kPa} \quad 2 \text{ marks}$$



$$p_{\text{eq}} \quad 1,080 \cdot 10^3 - 2x \quad x \quad 1 \text{ mark}$$

$$K_p^\circ = \frac{p_r(\text{N}_2\text{O}_4)}{p_r^2(\text{NO}_2)} = \frac{p(\text{N}_2\text{O}_4) \cdot p^\circ}{p^2(\text{NO}_2)} = \frac{x \cdot 100}{(1,081 \cdot 10^3 - 2x)^2} = 1,00 \quad 1 \text{ mark}$$

$$x = 436 \text{ kPa} \quad 2 \text{ marks}$$

$$p(\text{total}) = p(\text{NO}_2) + p(\text{N}_2\text{O}_4) = (1,08 \cdot 10^3 - 2x) + x = 1,081 \cdot 10^3 - x \quad 1 \text{ mark}$$

$$p(\text{total}) = 1081 - 436 = 645 \text{ kPa} \quad 2 \text{ marks}$$

Other solutions that lead to the same result are accepted.

9 marks

$$K_{\text{cr}}(\text{H}_2\text{O}) = 1,86 \text{ } ^\circ\text{C}\cdot\text{mol}^{-1}\cdot\text{kg}^{-1}$$

$$K_{\text{eb}}(\text{H}_2\text{O}) = 0,52 \text{ } ^\circ\text{C}\cdot\text{mol}^{-1}\cdot\text{kg}^{-1}$$

$$p^\circ(\text{H}_2\text{O}) = 31,7 \text{ hPa}$$

$$R = 0,08205 \text{ atm}\cdot\text{L}\cdot\text{K}^{-1}\cdot\text{mol}^{-1}$$

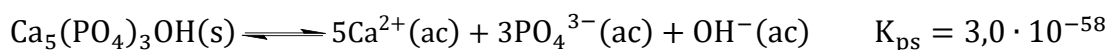
Substance	NO ₂	N ₂ O ₄
$\Delta H_f^\circ \text{ (kJ}\cdot\text{mol}^{-1}\text{)}$	33,18	9,16
$S^\circ \text{ (J}\cdot\text{K}^{-1}\cdot\text{mol}^{-1}\text{)}$	240,06	304,29

Problem 9. Apatite (25 points)

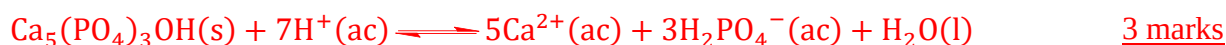
9.1	9.2	9.3	9.4	9.5	9.6	9.7	9.8	Total	Total
3	5	8	5	3	4	15	7	50	25
3	5	8	5	3	4	15	7	50	25

Among the minerals known as apatites, those with the simplest composition have the formula $\text{Ca}_5(\text{PO}_4)_3\text{X}$; the X^- anion can be Cl^- (chloroapatite), F^- (fluoroapatite) or OH^- (hydroxyapatite). Apatites are the main sources of phosphate, and therefore are essential in the manufacture of mineral fertilizers and obtaining phosphorus.

Hydroxyapatite is present in the bones and teeth of the human body. It constitutes about 60-70% of bone tissue, to which it gives great mechanical resistance; and contains 99% of the calcium and 80% of the total phosphorus. Hydroxyapatite is a mineral that is slightly soluble in water, so it has a very small solubility product.



9.1. Hydroxyapatite is dissolved by the action of mineral acids such as hydrochloric acid. State the ionic reaction that occurs; suppose that H_2PO_4^- is formed.



Hydrolysis of dental enamel hydroxyapatite causes the formation of caries. This process occurs when the bacteria present in the mouth metabolize the sugars in the remains of food and transform them into monoprotic acids such as acetic and lactic. When the pH of saliva is less than 5.5, hydroxyapatite begins to dissolve and tooth decay occurs. For this reason, it is necessary to maintain adequate oral hygiene.

9.2. Calculate the molar concentration of lactic acid required to bring the pH of saliva to 5.5.

The dissociation constant of lactic acid is high, so its dissociation cannot be neglected.

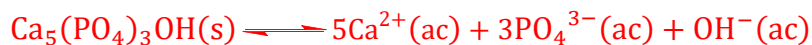
$$K_{\text{a}} = \frac{[\text{H}^+][\text{Lact}^-]}{[\text{HLact}]} \approx \frac{[\text{H}^+]^2}{c_0(\text{HLact}) - [\text{H}^+]} \quad \underline{\text{3 marks}}$$

$$c_0(\text{HLact}) = \frac{[\text{H}^+]^2}{K_{\text{a}}} + [\text{H}^+] = \frac{[10^{-5,5}]^2}{10^{-3,86}} + 10^{-5,5} = 3,2 \cdot 10^{-6} \text{ mol} \cdot \text{L}^{-1} \quad \underline{\text{5 marks}}$$

9.3. Show that the solubility S (in $\text{mol} \cdot \text{L}^{-1}$) of hydroxyapatite in water, as a function of pH and the corresponding equilibrium constants, obeys the following expression:

$$S = \sqrt[8]{\frac{K_{\text{ps}} \cdot 10^{-\text{pH}}}{84375 \cdot \alpha_3^3 \cdot K_{\text{w}}}} \quad \alpha_3 = \frac{[\text{PO}_4^{3-}]}{c(\text{PO}_4^{3-})_{\text{total}}} = \frac{K_{\text{a}1}K_{\text{a}2}K_{\text{a}3}}{[\text{H}^+]^3 + K_{\text{a}1}[\text{H}^+]^2 + K_{\text{a}1}K_{\text{a}2}[\text{H}^+] + K_{\text{a}1}K_{\text{a}2}K_{\text{a}3}}$$

Note: You do not have to prove the expression for α_3 .



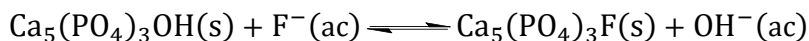
$$K_{\text{ps}} = [\text{Ca}^{2+}]^5 [\text{PO}_4^{3-}]^3 [\text{OH}^-] \quad \underline{\text{1 mark}}$$

$[\text{Ca}^{2+}] = 5 \cdot S$	$[\text{OH}^-] = K_w/10^{-\text{pH}}$	1 mark c/u	2 marks
$[\text{PO}_4^{3-}] = \alpha_3 \cdot c(\text{PO}_4^{3-})_{\text{total}} = \alpha_3 \cdot 3 \cdot S$			2 marks
$K_{\text{ps}} = [\text{Ca}^{2+}]^5 [\text{PO}_4^{3-}]^3 [\text{OH}^-] = (5 \cdot S)^5 \cdot (\alpha_3 \cdot 3 \cdot S)^3 \cdot K_w/10^{-\text{pH}}$			
$K_{\text{ps}} = 3125 \cdot S^5 \cdot 27 \cdot \alpha_3^3 \cdot S^3 \cdot K_w/10^{-\text{pH}} = 84375 \cdot \alpha_3^3 \cdot S^8 \cdot K_w/10^{-\text{pH}}$			2 marks
$S = \sqrt[8]{\frac{K_{\text{ps}} \cdot 10^{-\text{pH}}}{84375 \cdot \alpha_3^3 \cdot K_w}}$			1 mark
			<u>8 marks</u>

9.4. Calculate the solubility (in $\text{mol} \cdot \text{L}^{-1}$) of hydroxyapatite in water at $\text{pH} = 5.5$.

At $\text{pH} = 5.5$, the predominant dissolved phosphorus species are H_2PO_4^- and HPO_4^{2-} .	
$\alpha_3 \approx \frac{K_{a1} K_{a2} K_{a3}}{K_{a1} [\text{H}^+]^2 + K_{a1} K_{a2} [\text{H}^+]}$	1 mark
$\alpha_3 = \frac{10^{-2,12} 10^{-7,21} 10^{-12,67}}{10^{-2,12} (10^{-5,5})^2 + 10^{-2,12} 10^{-7,21} 10^{-5,5}} = 1,29 \cdot 10^{-9}$	2 marks
$S = \sqrt[8]{\frac{K_{\text{ps}} \cdot 10^{-\text{pH}}}{84375 \cdot \alpha_3^3 \cdot K_w}} = \sqrt[8]{\frac{3,0 \cdot 10^{-58} \cdot 10^{-5,5}}{84375 \cdot \alpha_3^3 \cdot 1,0 \cdot 10^{-14}}} = 3,9 \cdot 10^{-4} \text{ mol} \cdot \text{L}^{-1}$	2 marks
	<u>5 marks</u>

Many commercial mouthwashes and toothpastes add fluoride to their formulation to help prevent cavities. However, excess fluoride is harmful, so ingestion of these preparations should be avoided and use in young children should be controlled. The fluoride ion is capable of stopping tooth decay because it allows the formation of fluoroapatite, which is less soluble ($K_{\text{ps}} = 7.1 \cdot 10^{-61}$) than hydroxyapatite, according to the reaction



9.5. Calculate the equilibrium constant for the above reaction.

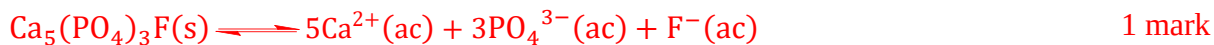
$K_e = \frac{K_{\text{ps}}(\text{hydroxyapatite})}{K_{\text{ps}}(\text{fluoroapatite})} = \frac{3,0 \cdot 10^{-58}}{7,1 \cdot 10^{-61}} = 423$	<u>3 marks</u>
--	----------------

9.6. Calculate the concentration of fluoride ions required for fluoroapatite to form at $\text{pH} 5.5$. If you could not solve part 9.5, consider $K = 500$.

$[\text{OH}^-] = 10^{-\text{pOH}} = 10^{-14+\text{pH}} = 10^{-14+5,5} = 10^{-8,5} = 3,16 \cdot 10^{-9} \text{ mol} \cdot \text{L}^{-1}$	2 marks
$K_e = \frac{[\text{OH}^-]}{[\text{F}^-]} \rightarrow [\text{F}^-] = \frac{[\text{OH}^-]}{K_e} = \frac{3,16 \cdot 10^{-9}}{423} = 7,48 \cdot 10^{-12} \text{ mol} \cdot \text{L}^{-1}$	2 marks
	<u>4 marks</u>

Utilizando el valor de $K = 500$, se obtiene $6,32 \cdot 10^{-12} \text{ mol} \cdot \text{L}^{-1}$.

9.7. Calculate the solubility of fluoroapatite in water at $\text{pH} = 5.5$.



$$K_{\text{ps}} = [\text{Ca}^{2+}]^5 [\text{PO}_4^{3-}]^3 [\text{F}^-] \quad 1 \text{ mark}$$

$$[\text{Ca}^{2+}] = 5 \cdot S \quad [\text{F}^-] = \alpha_1 \cdot c(\text{F}^-)_{\text{total}} = \alpha_1 \cdot S \quad 1 \text{ mark c/u} \quad 2 \text{ marks}$$

$$[\text{PO}_4^{3-}] = \alpha_3 \cdot c(\text{PO}_4^{3-})_{\text{total}} = \alpha_3 \cdot 3 \cdot S \quad 2 \text{ marks}$$

$$K_{\text{ps}} = [\text{Ca}^{2+}]^5 [\text{PO}_4^{3-}]^3 [\text{OH}^-] = (5 \cdot S)^5 \cdot (\alpha_3 \cdot 3 \cdot S)^3 \cdot \alpha_1 \cdot S \quad 1 \text{ mark}$$

$$K_{\text{ps}} = 3125 \cdot S^5 \cdot 27 \cdot \alpha_3^3 \cdot S^3 \cdot \alpha_1 \cdot S = 84375 \cdot \alpha_3^3 \cdot \alpha_1 \cdot S^9 \quad 1 \text{ mark}$$

$$S = \sqrt[9]{\frac{K_{\text{ps}}}{84375 \cdot \alpha_3^3 \cdot \alpha_1}} \quad 2 \text{ marks}$$

At $\text{pH} = 5,5$, the predominant phosphorus species are H_2PO_4^- and HPO_4^{2-} . 1 mark

$$\alpha_3 \approx \frac{K_{a1}K_{a2}K_{a3}}{K_{a1}[\text{H}^+]^2 + K_{a1}K_{a2}[\text{H}^+]} = 1,29 \cdot 10^{-9} \quad 1 \text{ mark}$$

At $\text{pH} = 5,5$, the predominant fluorine species is F^- and $\alpha_1 \approx 1,00$. 2 marks

$$S = \sqrt[9]{\frac{7,1 \cdot 10^{-61}}{84375 \cdot (1,29 \cdot 10^{-9})^3 \cdot 1,00}} = 5,40 \cdot 10^{-5} \text{ mol} \cdot \text{L}^{-1} \quad 1 \text{ mark}$$

15 marks

9.8. Is there solid calcium fluoride ($K_{\text{ps}} = 3,6 \cdot 10^{-11}$) in equilibrium with the fluorapatite under the above conditions? If you were unable to solve part 9.7, consider $S = 1,00 \cdot 10^{-5} \text{ mol} \cdot \text{L}^{-1}$.

$$[\text{Ca}^{2+}] = 5 \cdot S = 5 \cdot 5,40 \cdot 10^{-5} = 2,70 \cdot 10^{-4} \text{ mol} \cdot \text{L}^{-1} \quad 2 \text{ marks}$$

$$[\text{F}^-] = \alpha_1 \cdot S = 5,40 \cdot 10^{-5} \text{ mol} \cdot \text{L}^{-1} \quad 2 \text{ marks}$$

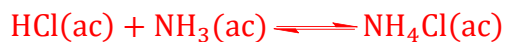
$$[\text{Ca}^{2+}][\text{F}^-]^2 = (2,70 \cdot 10^{-4})(5,40 \cdot 10^{-5})^2 = 7,87 \cdot 10^{-13} \quad 2 \text{ marks}$$

Since $PI < K_{\text{sp}}$, no solid calcium fluoride is formed. 1 marks 7 marks

acid	acetic	lactic	HF	H_3PO_4
pK_a	4,76	3,86	3,17	2,12; 7,21; 12,67

$$\text{pX} \equiv -\log X$$

10.4. Calculate the pH of the solution resulting from mixing 10 mL of $0.60 \text{ mol} \cdot \text{L}^{-1}$ hydrochloric acid with 20 mL of $0.50 \text{ mol} \cdot \text{L}^{-1}$ ammonia. $K_b(\text{NH}_3) = 1,8 \cdot 10^{-5}$



$$n(\text{HCl}) = c(\text{HCl}) \cdot V(\text{HCl}) = 0,60 \cdot 10 = 6,0 \text{ mmol} \quad 1 \text{ mark}$$

$$n(\text{NH}_3) = c(\text{NH}_3) \cdot V(\text{NH}_3) = 0,50 \cdot 20 = 10,0 \text{ mmol} \quad 1 \text{ mark}$$

$$n(\text{NH}_3)_{\text{exc}} = 10,0 - 6,0 = 4,0 \text{ mmol} \quad 1 \text{ mark}$$

$$n(\text{NH}_4\text{Cl}) = n(\text{HCl}) = 6,0 \text{ mmol} \quad 1 \text{ mark}$$



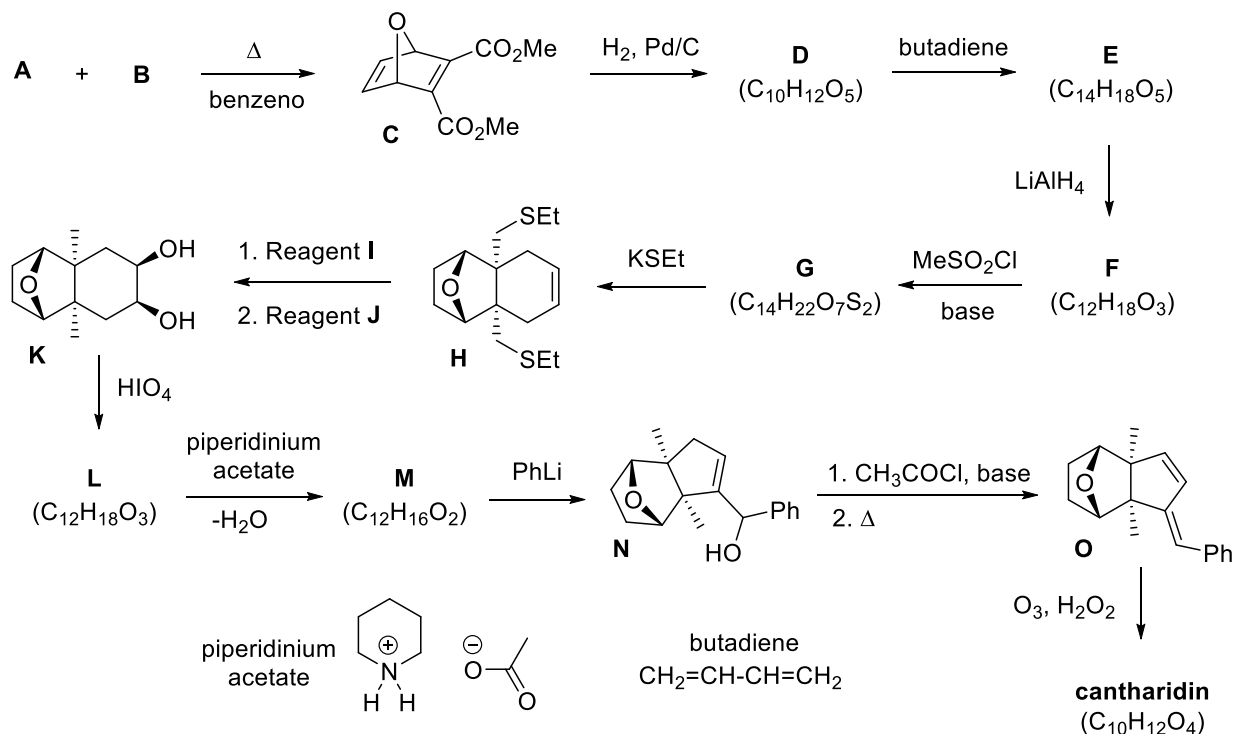
$$[\text{OH}^-] = K_b \cdot \frac{[\text{NH}_3]}{[\text{NH}_4^+]} = 1,8 \cdot 10^{-5} \cdot \frac{4,0/30}{6,0/30} = 1,2 \cdot 10^{-5} \text{ mol} \cdot \text{L}^{-1} \quad 1 \text{ mark}$$

$$\text{pH} = 14 - \text{pOH} = 14 + \log(1,2 \cdot 10^{-5}) \approx 9,1 \quad 1 \text{ mark} \quad \underline{6 \text{ marks}}$$

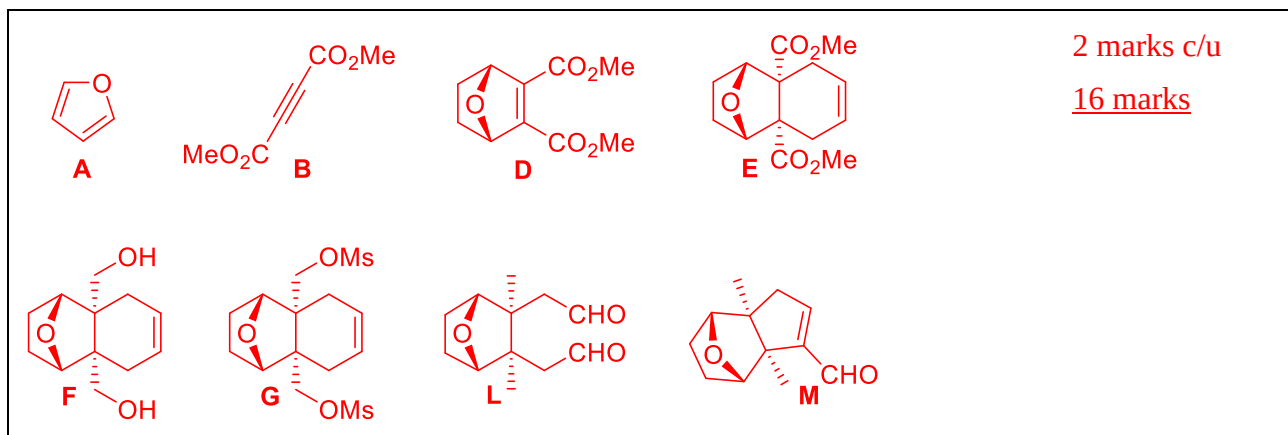
Problem 11. Cantharidin (19 points)

11.1	11.2	11.3	11.4	11.5	11.6	11.7	Total	Total
16	4	4	4	2	2	4	38	19
16	4	4	4	2	2	4	38	19

Cantharidin is a poisonous chemical compound produced by beetles of the Meloidae family. It was first isolated by the French chemist Pierre Jean Robiquet in 1812. It can be obtained starting from the Diels-Alder reaction between diene **A** and dienophile **B**, as shown below (Stork, G. *et al*, *J. Am. Chem. Soc.* **1951**, 73, 4501; Stork, G. *et al*, *J. Am. Chem. Soc.* **1953**, 75, 384-392).



11.1. Identify compounds **A**, **B**, **D**, **E**, **F**, **G**, **L** and **M**.



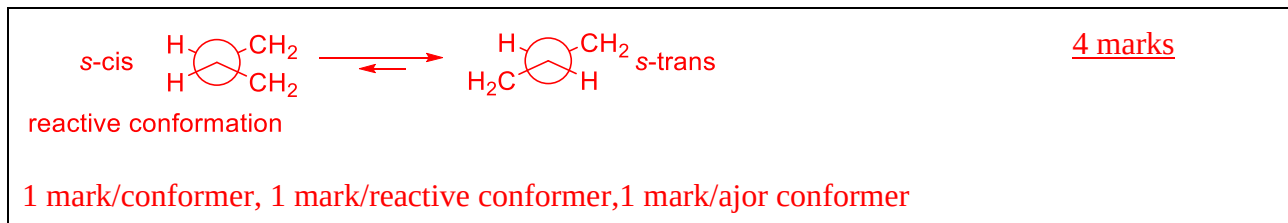
2 marks c/u

16 marks

11.2. Draw the structure of cantharidin. It is known that its structure is symmetrical, it does not add bromine water and it reacts with amines and hydrazines.



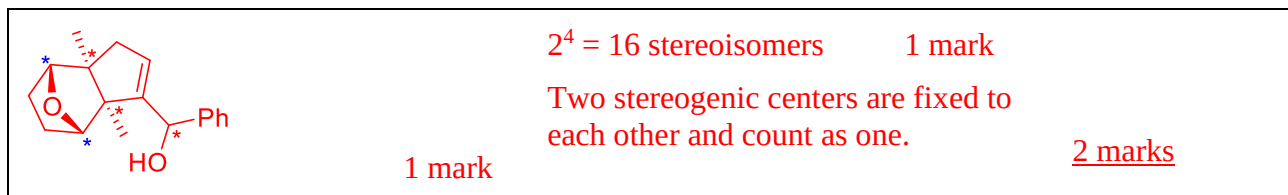
11.3. State the balance between the s-cis and s-trans conformations around the central single bond of butadiene; Indicate which of the two forms is found in greater proportion. Use Newman projections to represent the conformers. Say which of them is the one that participates in the cycloaddition reaction.



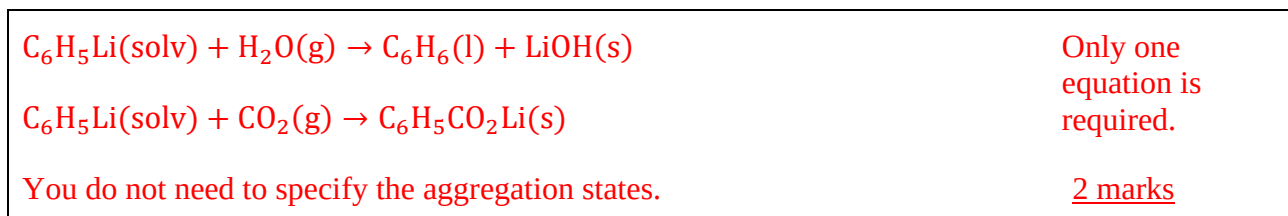
11.4. Suggest the reagents used to transform **H** into **K**. Reagent **I** makes it possible to hydroxylate the alkene present in the tricycle **H**; while reagent **J** allows to reduce the C-S bond.



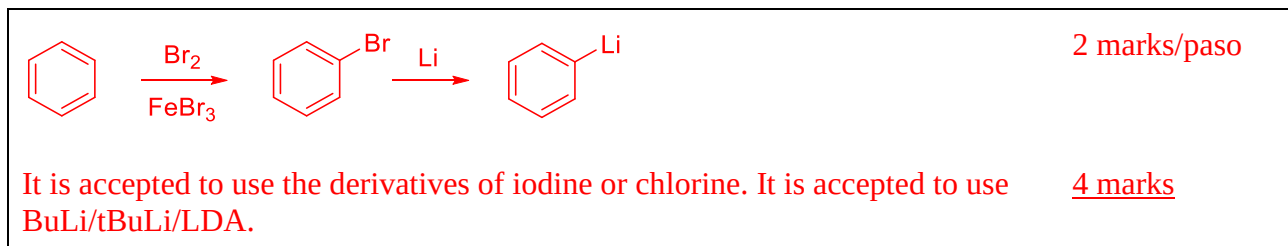
11.5. Label the stereogenic centers of compound **N**. How many stereoisomers are theoretically possible?



11.6. To use phenyllithium it is necessary to maintain an inert atmosphere. State one of the possible side reactions that can occur if this precaution is not taken into account.



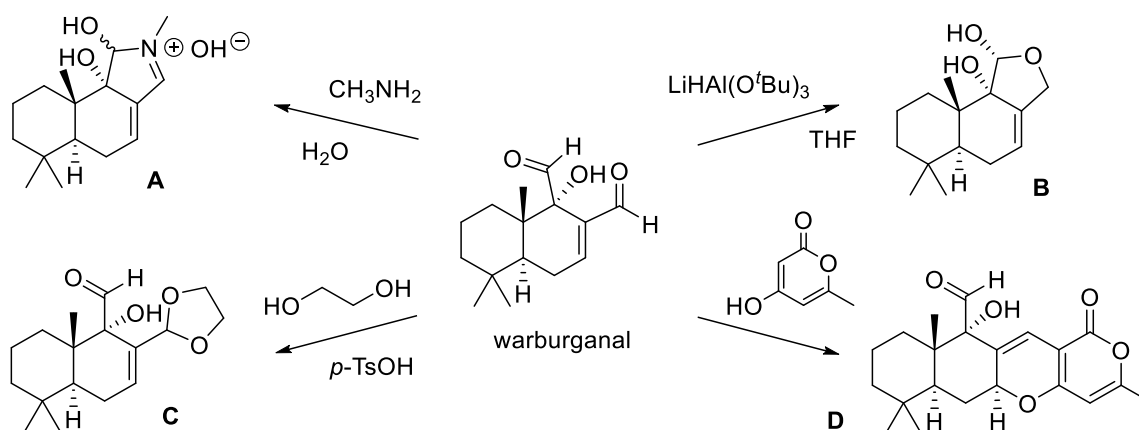
11.7. Propose a way to obtain phenyllithium (PhLi) using benzene as the only organic starting product.



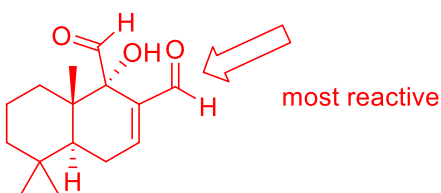
Problem 12. Thuajopsene, warburganal and shikimic acid (21 points)

12.1	12.2	12.3	12.4	12.5	12.6	12.7	12.8
2	7	2	2	2	4	4	2
2	7	2	2	2	4	4	2
12.9	12.10	12.11	12.12	12.13	12.14	Total	Total
2	2	4	4	2	3	42	21
2	2	4	4	2	3	42	21

Semi-synthesis or partial synthesis consists of chemically obtaining compounds of interest starting from a natural product that has not been previously synthesized, but rather was extracted and purified from a microorganism using mixture separation techniques. It is used in industry and in research when it is a better alternative to a total synthesis, generally taking into account time and cost criteria. Warburganal, thujopsene and shikimic acid are three natural products that have been used in partial synthesis processes. Below are examples of chemical modifications that can be made to these substances.



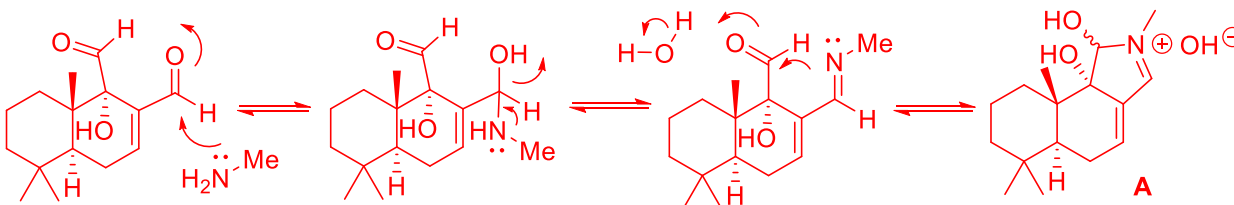
12.1. In the structure of warburganal, point to the most reactive carbonyl group. How do you explain the difference in reactivity between like functional groups in the molecule?

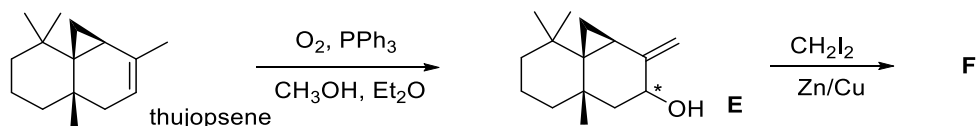
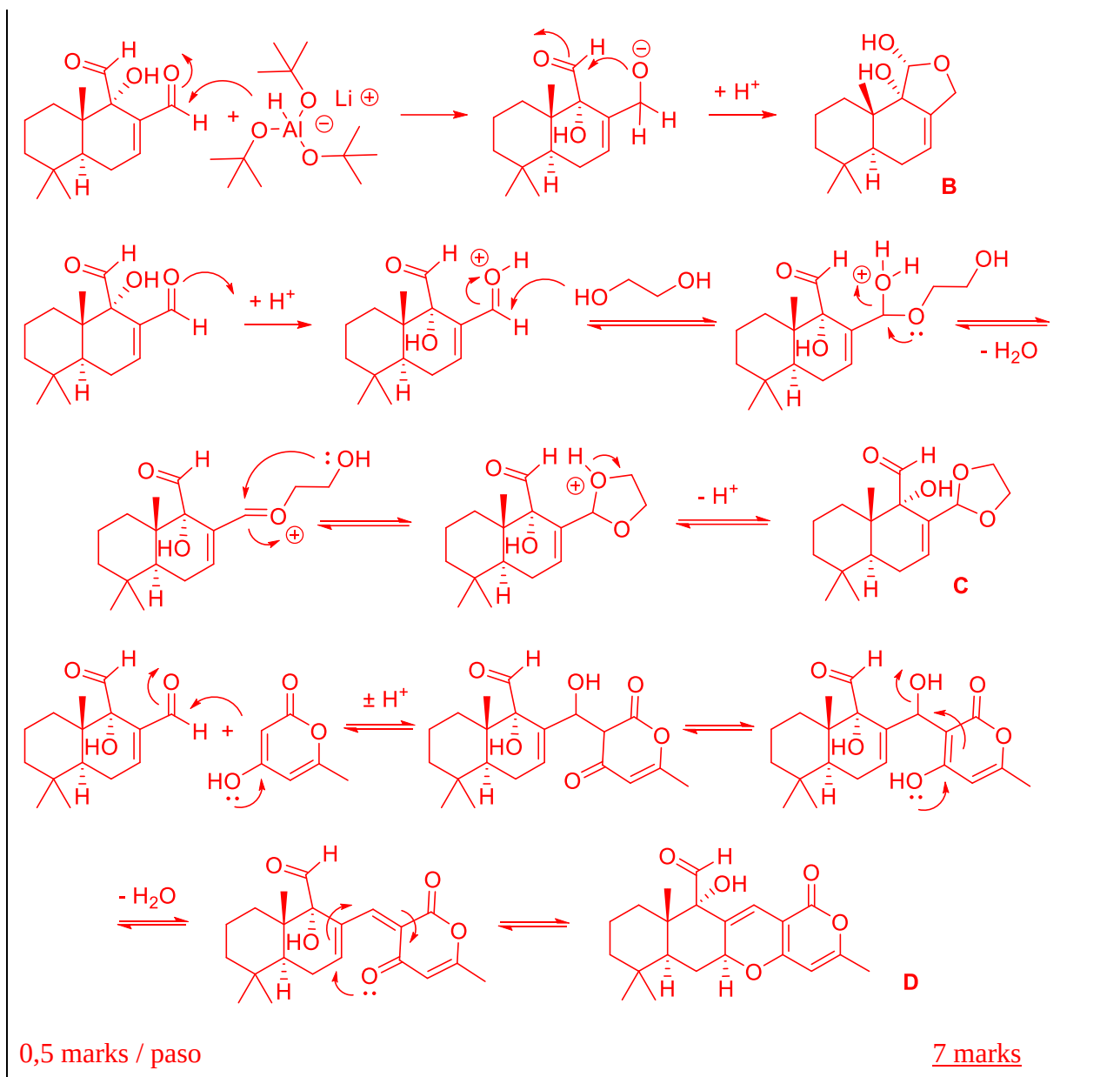


indicated aldehyde group presents less steric hindrance.

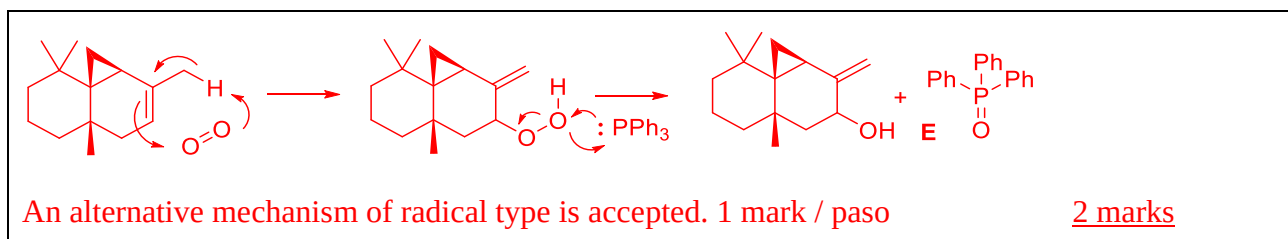
2 marks

12.2. Propose mechanisms for all the reactions represented in the schematic. *p*-TsOH: *para*-methylbenzenesulfonic acid.





12.3. Thujopsene, in the presence of oxygen from the air and triphenylphosphine, is oxidized to allyl alcohol **E**. Propose a mechanism for the reaction described.



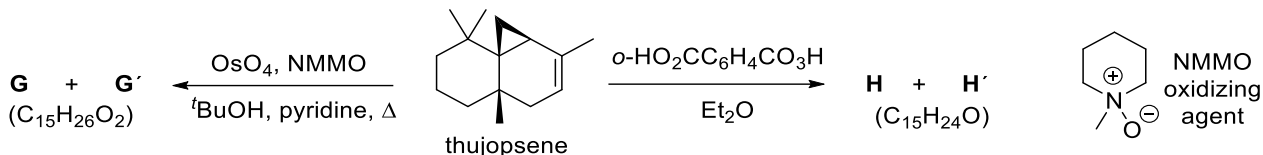
12.4. Which stereoisomer of alcohol **E** do you think is the major product in the above reaction? Represent the corresponding structure.



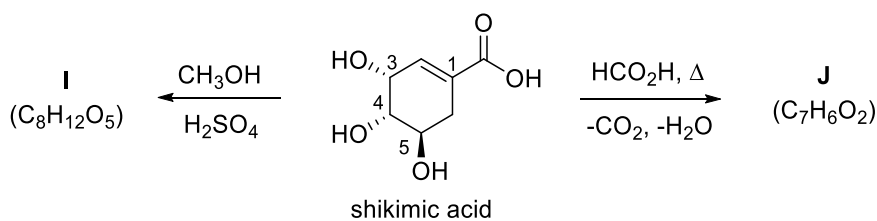
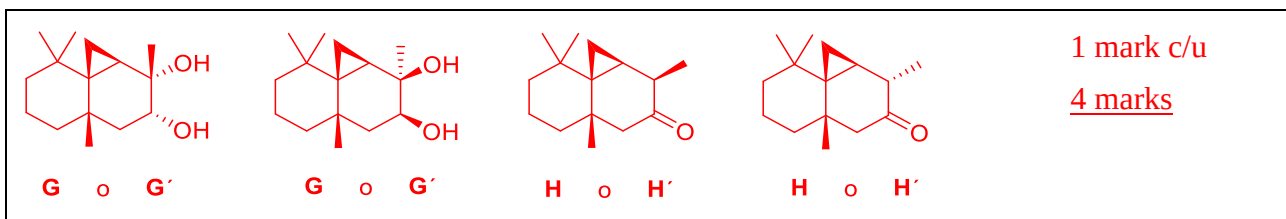
12.5. Suppose **E** (the isomer you found in 12.4) is reacted with the Simmons-Smith reagent. Represent the structure of the major product **F** that is obtained; note the stereochemistry.



Product of the presence of a double bond in its structure, thujopsene reacts with OsO_4 and *o*-peroxybenzoic acid, as shown below.



12.6. Represent the structures of **G** and **H**, and their respective stereoisomers **G'** and **H'**. It is known that **H** and **H'** are not the expected products of the reaction of thujopsene with a peracid: both have only four asymmetric carbons in their structure and react with 2,4-dinitrophenylhydrazine.



12.7. Represent the structures of **I** and **J**.

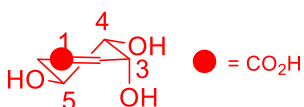
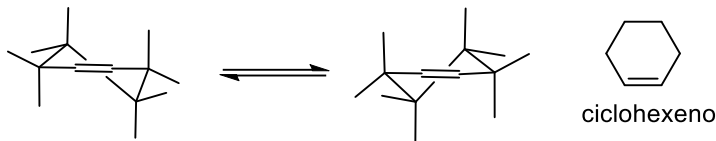


12.8. How could you selectively protect the hydroxyl groups that occupy positions 3 and 4 in the shikimic acid molecule?

By formation of the corresponding (a)ketal, for example, with acetone.

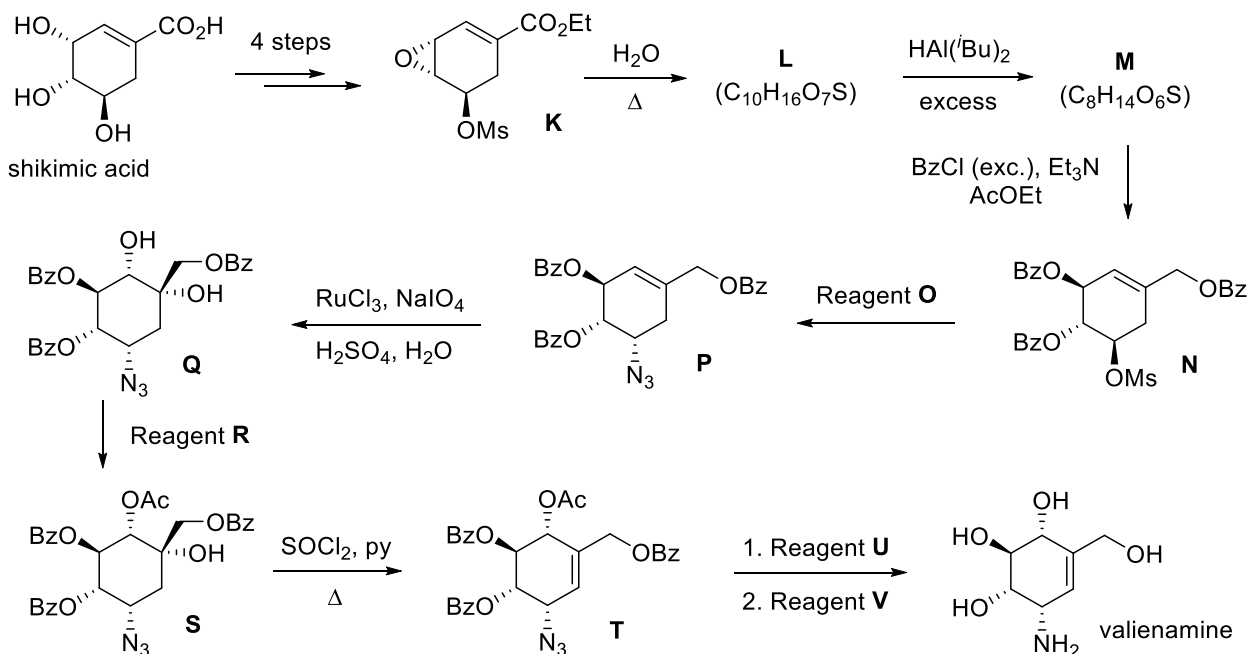
2 marks

12.9. Draw the most stable conformation for shikimic acid. Assume that the molecule is in the form of a semi-chair, like cyclohexene. Show only the substituents that are not hydrogen atoms.



2 marks

Shikimic acid can be used as a starting material for the synthesis of aminocyclitol valienamine. This compound is found as part of several pseudo-oligosaccharides such as the antidiabetic drug acarbose and the antibiotic validamycin.

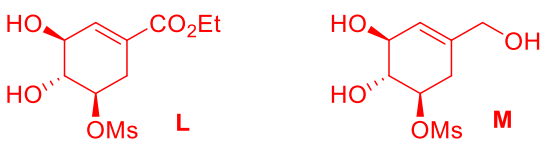


12.10. Name the compound valienamine according to IUPAC nomenclature.

(1S,2S,3R,6S)-6-amine-4-(hydroxymethyl)cyclohex-4-ene-1,2,3-triol

2 marks

12.11. Represent the structures of L and M. Ms: methanesulfonyl or mesyl, $-\text{SO}_2\text{CH}_3$.



2 marks c/u

4 marks

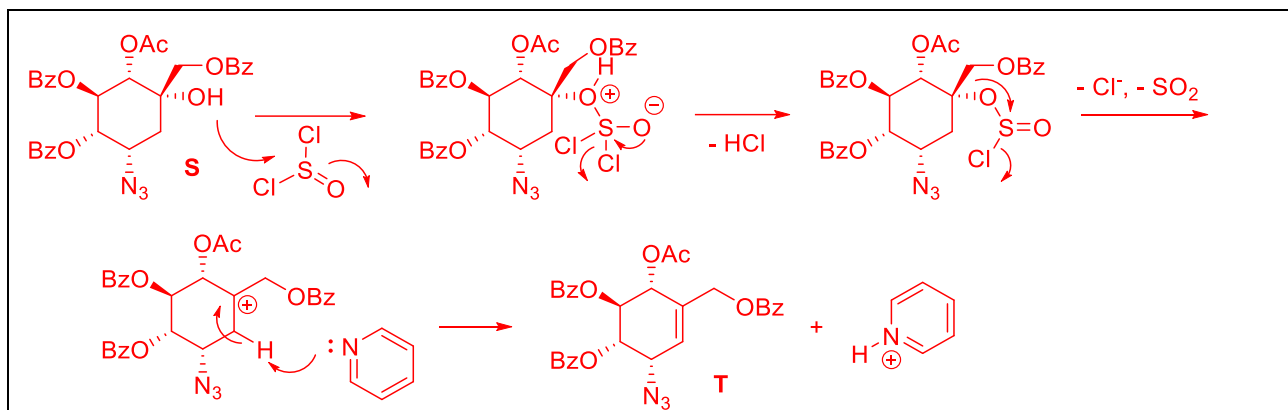
12.12. Suggest the possible reagents identified in the scheme as **O**, **R**, **U**, and **V**. It is known that **U** allows the hydrolysis of the acetyl (Ac) and benzoyl (Bz) groups, while **V** allows the azide group to be reduced to amino.

O: NaN₃ **R:** (AcO)₂O **U:** NaOH/MeOH, K₂CO₃ **V:** PPh₃/H₂, Pd/LiAlH₄

1 mark c/u

4 marks

12.13. State a mechanism for the transformation of **S** into **T**.



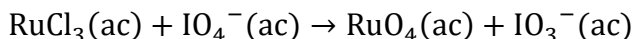
0,5 marks each step.

It is accepted to suggest that the last two steps are concerted.

2 marks

In the above sequences, osmium(VIII) and ruthenium(VIII) oxides are used to dihydroxylate thujopsene and alkene **P**, respectively. As osmium and ruthenium compounds are very toxic and expensive, one way to carry out this transformation is to generate these reagents in situ by the action of oxidizing agents such as N-methylmorpholine oxide (NMMO) or potassium periodate.

12.14. Fit the chemical equation for the reaction for the in situ formation of ruthenium(VIII) oxide from ruthenium(III) chloride by the ion-electron method.



1 mark c/half-reaction 2 marks total equation

3 marks

Note: Alternative reaction mechanisms are accepted, provided they are logical and important intermediaries are not omitted. It is accepted that the proton transfer steps are omitted, except in the acetal formation mechanism (question 12.2.).